

Q1a - Graduate Paper Analysis

Pedro Fischetti | CSCE 580 Finals

a) Paper & Presenter

Paper: "Towards Enhancing Road Safety in South Carolina Using Insights from Traffic & Driver-Education Data"

Presented by: Sai Teja Paladi

Published at AAAI-25 in Philadelphia. The research team included folks from USC, SC Dept of Public Safety, National Safety Council, and DHEC.

b) My Example

The big takeaway from this paper is that you can use data to figure out *where* accidents happen most, *why* they happen (road type, lack of education), and then make targeted fixes instead of just throwing money at the problem randomly.

I thought about how this would work if we applied it to **USC's campus**.

The Setup

Say we collected:

- Campus police reports on parking lot fender-benders and pedestrian incidents (2019-2024)
- Shuttle bus data showing where traffic gets congested
- Records of which students took any kind of driving safety orientation

What We'd Probably Find

Problem spots (like Greenville/Charleston in the paper):

- Assembly Street by the stadium - crazy traffic on game days and just in general
- Greene Street crossings - students dart across between classes all the time
- Those parking garage exits where visibility is terrible

Infrastructure issues (like two-way undivided roads):

- Sumter Street has no real barrier between cars and pedestrians
- The Bull Street crossings by Greek Village don't have protected signals
- Five Points is a mess of jaywalkers and cars

Education gaps (like Alive@25 declining in rural areas):

- Most students never take any campus driving course - it's optional and nobody bothers
- International students often don't know US traffic rules/norms
- Freshmen have the most parking lot incidents (no surprise there)

What We'd Recommend

Same logic as the paper:

- Put up protected crossings on Greene Street
- Add barriers on Assembly Street's median
- Make freshmen and international students do a quick driving orientation before they can register a car on campus

c) Why This Supports the Paper's Conclusions

The paper basically makes a few key points, and my campus example hits all of them:

1. Accidents cluster in specific places

The paper found urban counties (Greenville, Charleston, Richland) have way more crashes than rural ones. On campus, you'd see the same thing - Assembly Street and Greene Street would dominate the incident reports while random side streets would be fine. The point is: don't spread resources thin, focus on the hot spots.

2. Road design matters a lot

Their data showed two-way undivided roads are the most dangerous. On campus, streets without physical barriers or protected crossings would show the same pattern. It's not rocket science - if there's nothing separating pedestrians from cars, bad things happen more often.

3. Education programs help, but only if people actually use them

The paper found Alive@25 participation dropping in rural areas while accidents went up. On campus, optional safety courses get ignored. Making them mandatory (or at least strongly incentivized) for high-risk groups would probably move the needle.

4. You need different departments working together

The paper involved SCDOT, DHEC, National Safety Council, and academic researchers all sharing data. A campus version would need USC Police, Facilities, Student Affairs, and the city of Columbia to actually talk to each other - which is easier said than done, but that's kinda the point.

5. Recommendations should be specific, not vague

The paper doesn't just say "make roads safer" - it says prioritize Greenville, Charleston, and Richland for divided roads with barriers. A campus analysis should do the same: "fix the Greene Street crossing first" rather than "improve pedestrian safety."

Bottom Line

The paper's approach - use real data to find patterns, identify root causes, and make specific recommendations - works for state-level traffic safety and would work just as well for a university campus. It's basically the same methodology scaled down to a smaller area.